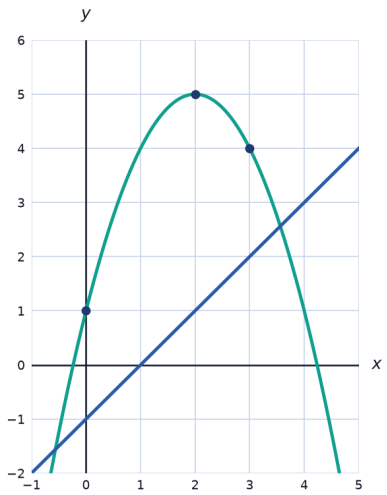
Trigonometric Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Graphing tangent functions

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(5)$.

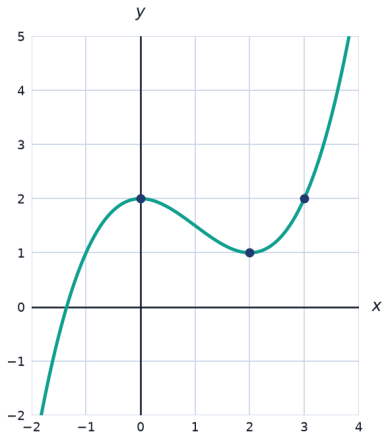


2 [Foundational] • Target: Graphing tangent functions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

3 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(2)$.



4 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

5 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

A data set recorded for key features of secant, cosecant, and cotangent graphs consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

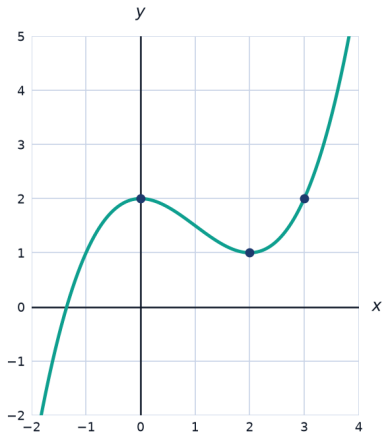
© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(3)$.

7 [Foundational] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

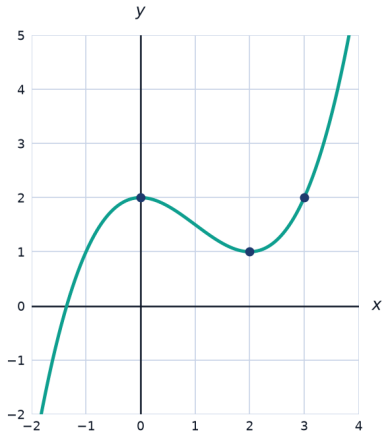
For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(4)$.

9 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

10 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

For the model $f(x) = 5x + 3$ in key features of secant, cosecant, and cotangent graphs, evaluate $f(5)$.



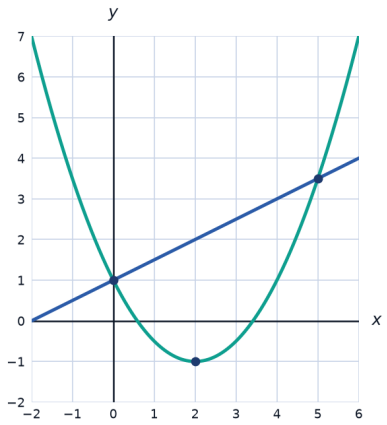
© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • Target: Key features of secant, cosecant, and cotangent graphs

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

12 [Intermediate] • Target: Graphing secant, cosecant, and cotangent functions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(2)$.



13 [Intermediate] • Target: Graphing secant, cosecant, and cotangent functions

A data set recorded for graphing secant, cosecant, and cotangent functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Graphing secant, cosecant, and cotangent functions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(3)$.

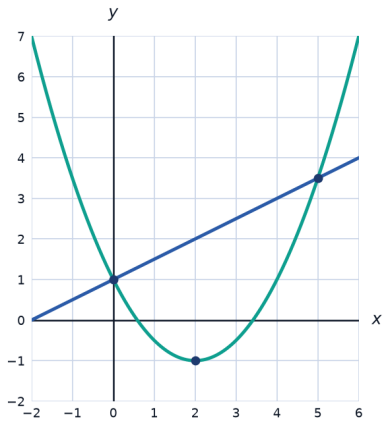
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphing secant, cosecant, and cotangent functions • Cross-Lesson: 3.01 Polynomial Functions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(4)$.

16 [Challenging] • Target: Graphing secant, cosecant, and cotangent functions • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 6x + 4$ in graphing secant, cosecant, and cotangent functions, evaluate $f(5)$.



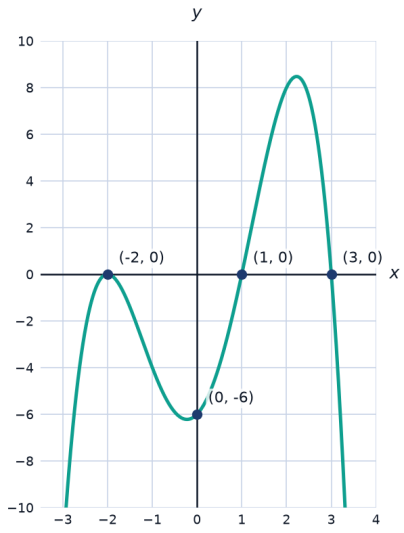
© ClipSAT — clipsat.org | All Rights Reserved

17 [Challenging] • Target: Graphing sine and cosine functions • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The cosine function is symmetric about the y-axis because it is:.

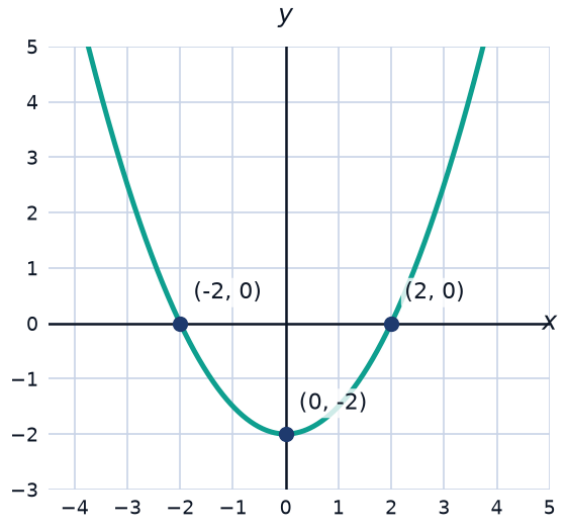
18 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 6.01 Radicals & Rational Exponents

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$? Show your complete algebraic work.



19 [Challenging] • Target: Key features of secant, cosecant, and cotangent graphs • Cross-Lesson: 7.01 Sequences & Series

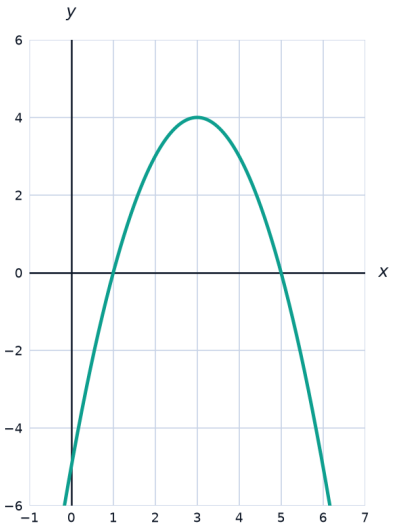
The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have? Show your complete algebraic work.



© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Graphing secant, cosecant, and cotangent functions • Cross-Lesson: 8.01 Conic Sections

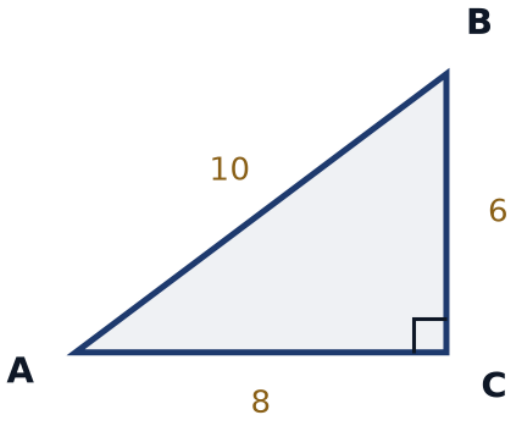
The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$? Show your complete algebraic work.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(2)$.



a. 11

b. 13

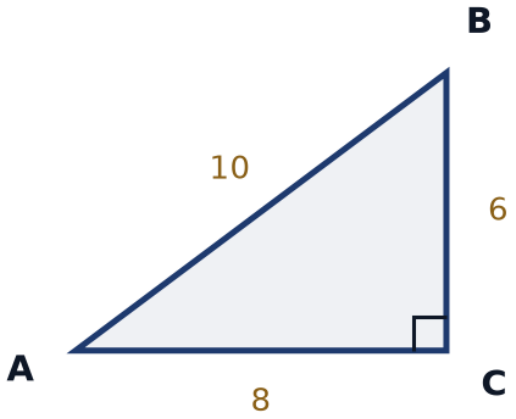
c. 4

d. 15

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

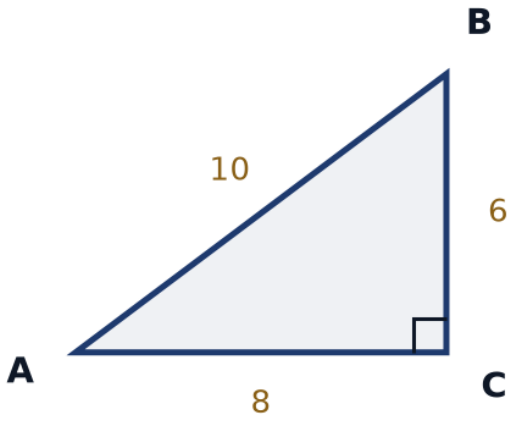
b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

23 [Foundational] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(5)$.



a. 17

b. 19

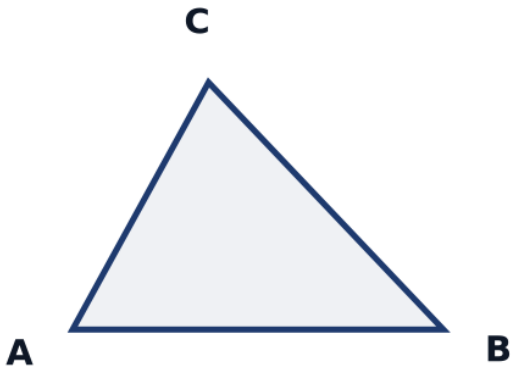
c. 10

d. 21

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Graphing sine and cosine functions

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(2)$.



a. 14

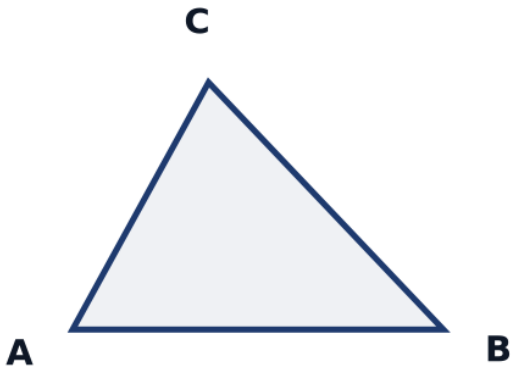
b. 17

c. 6

d. 18

25 [Foundational] • Target: Graphing sine and cosine functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

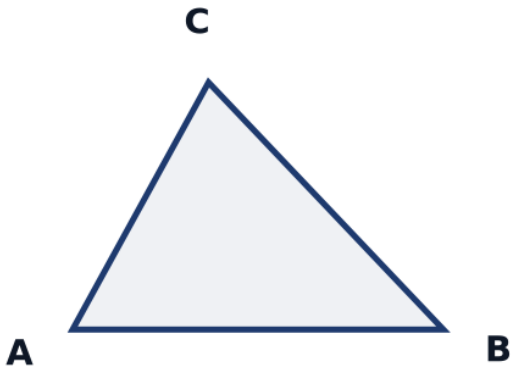
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Graphing sine and cosine functions

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(5)$.



a. 23

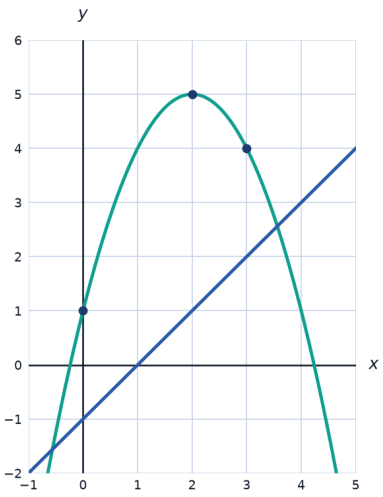
b. 26

c. 15

d. 27

27 [Foundational] • Target: Graphing tangent functions

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(2)$.



a. 17

b. 21

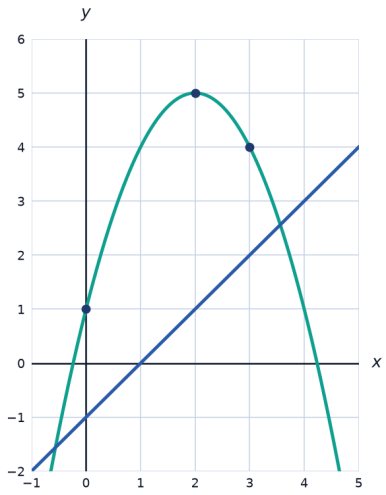
c. 8

d. 21

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Graphing tangent functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

A data set recorded for key features of sine, cosine, and tangent graphs consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(3)$.

a. 13

b. 15

c. 6

d. 17

32 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

For the model $f(x) = 2x + 7$ in key features of sine, cosine, and tangent graphs, evaluate $f(4)$.

a. 15

b. 17

c. 8

d. 19

33 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

34 [Intermediate] • Target: Key features of sine, cosine, and tangent graphs

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

35 [Intermediate] • Target: Graphing sine and cosine functions

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Graphing sine and cosine functions

A data set recorded for graphing sine and cosine functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

37 [Intermediate] • Target: Graphing sine and cosine functions

For the model $f(x) = 3x + 8$ in graphing sine and cosine functions, evaluate $f(3)$.

a. 17

b. 20

c. 9

d. 21

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Graphing sine and cosine functions • Cross-Lesson: 6.01 Radicals & Rational Exponents

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

39 [Challenging] • Target: Graphing sine and cosine functions • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

40 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 8.01 Conic Sections

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 1.01 Functions & Their Graphs

A data set recorded for graphing tangent functions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

42 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(3)$.

a. 21

b. 25

c. 12

d. 25

43 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 11.01 Probability & Statistics

For the model $f(x) = 4x + 9$ in graphing tangent functions, evaluate $f(4)$.

a. 25

b. 29

c. 16

d. 29

44 [Challenging] • Target: Graphing tangent functions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

45 [Challenging] • Target: Key features of secant, cosecant, and cotangent graphs • Cross-Lesson: 3.01 Polynomial Functions

A data set recorded for key features of secant, cosecant, and cotangent graphs consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

© ClipSAT — clipsat.org | All Rights Reserved